

Comparing a Real Spring to an Ideal Spring

Sage Buesing

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Using Δy as our difference between the spring's equilibrium length and its extended length, we can set up the force equation in the y direction for the hanging object.

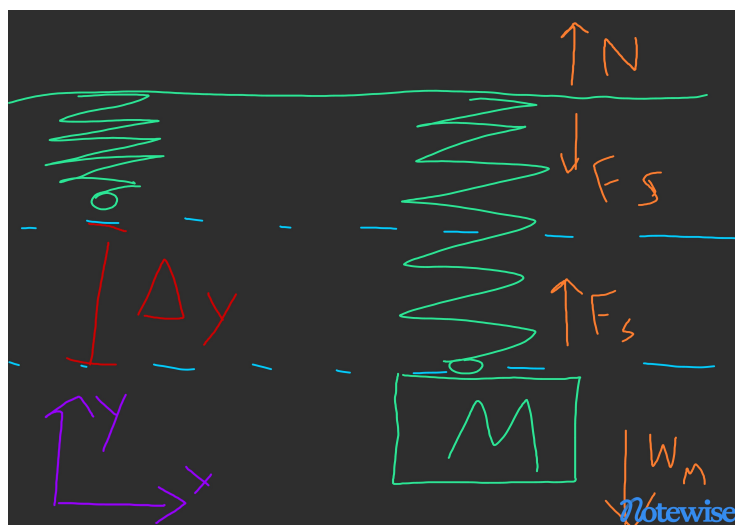


Figure 1: Hanging spring without a load (left) and load-bearing (right).

The net force must be 0 (in all directions) since we are considering the static/equilibrium state. I will suppress adding the y label, since we know all forces are in the y direction anyways

$$\begin{aligned} F_{NET} &= F_s - W_M = Ma = 0 \\ &\implies \\ F_s &= W_M \end{aligned} \tag{1}$$

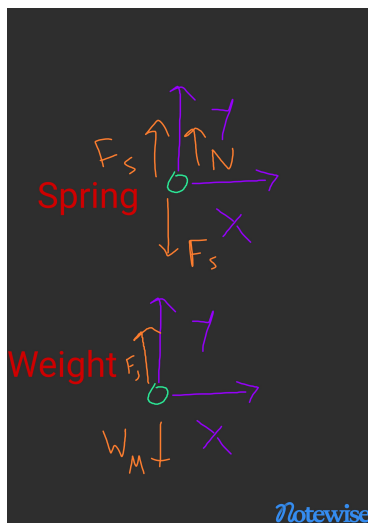


Figure 2: (Above) The forces on the spring. (Below) The forces on the weight.

We can then use the fact that the force of a stretched spring is given by $F_s = -k\Delta y$

$$-k\Delta y = W_m \quad (2)$$

$$\Delta y = -\frac{W_m}{k} \quad (3)$$

Note the minus sign, which indicates that the direction that the spring stretches will be down, or in the $-y$ direction.

We can plot this linear equation relating the weight to the spring displacement (see below). The slope of this line corresponds to k , the spring constant. k is a measure of how "stiff" the spring is.

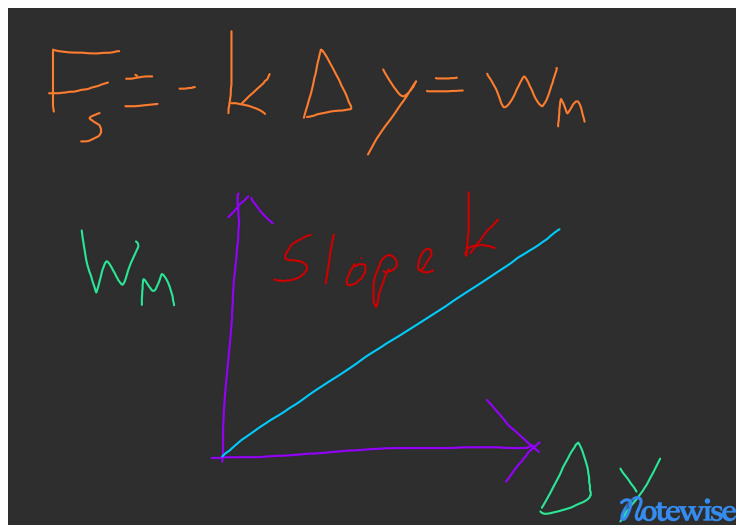


Figure 3: Plotting the (magnitude of) spring force (which is equal to the weight of the load) versus the extension of the spring.